IEDA 1250, Summer 2026

Final Exam Marking Rubric

Total: 120 points

General marking guidance

- Award full credit for any mathematically equivalent method with correct reasoning, even when notation differs from the official solution.
- Accept exact fractions or sufficiently accurate decimals. Minor rounding differences should not be penalized.
- Apply follow-through credit: after one arithmetic error, grade later steps using the student's resulting value when the method remains correct. Do not penalize the same error twice.
- Unsupported final answers receive only the points explicitly assigned to the final result; requested graphical or dynamic-programming work must be shown.
- For Questions 6 and 7, use the formula printed in the Summer 2026 paper, whose first denominator is $\lambda(\lambda + \mu_1)$.
- For Question 7, accept any valid stable numerical instance that proves the claimed strict comparison; the official instance is only one example.

Answer-key summary

Question	Key result
1	Player 1: $(3/8, 5/8)$; Player 2: $(1/2, 1/2)$; game value to Player 1: $1/2$.
2	$p^* = (0, 1/3, 2/3, 0, 0, 0)$, $q^* = (1/2, 1/2, 0, 0, 0, 0)$, $v = 6$; the two stated LPs are primal and dual.
3	Study-day allocation $(2, 1, 3, 1)$; maximum grade points 23.
4	Location sequence (A, B, C, B, B); maximum net profit 37.
5	$\pi = (16, 8, 4, 2, 1)/31$, utilization $15/31$, loss $1/31$, $\mathbb{E}[N] = 26/31$, $\lambda_e = 60/31$, $\mathbb{E}[T] = 13/30$; capacity wins in (g), speed wins in (h).
6	$\mathbb{E}[N] = 1/[A(1-\rho)^2]$.
7	One valid instance: $(\lambda, \mu_1, \mu_2) = (5, 1, 10)$, giving $\mathbb{E}[N]_2 = 1331/978 > 1 = \mathbb{E}[N]_{\text{fast only}}$.

Question 1 (20 points)

Item	Expected work	Points
1(a)	Defines $x = \Pr(\text{Player 1 uses } S_1)$ and obtains both lines $L_1(x) = 4x - 1$ and $L_2(x) = 2 - 4x$.	3
1(a)	Correct graphical interpretation: Player 1 maximizes the lower envelope of the two payoff lines.	3

1(a)	Finds the intersection $x = 3/8$, value $v = 1/2$, and reports Player 1's strategy $(3/8, 5/8)$.	3
1(a)	Makes Player 1 indifferent and obtains Player 2's strategy $(1/2, 1/2)$.	3
1(b)	Constructs Player 2's correctly oriented payoff table $\left(\begin{smallmatrix} -3 & 1 \\ 2 & -2 \end{smallmatrix}\right)$.	3
1(b)	Obtains $H_1(q) = 2 - 5q$, $H_2(q) = 3q - 2$ and shows the graphical intersection at $(q, v_2) = (1/2, -1/2)$.	3
1(b)	Recovers $x = 3/8$ and explicitly checks the sign relationship $v_2 = -v_1 = -1/2$.	2
	Question 1 total	**20**

Graphing note: If the student gives correct algebra but no graph or envelope interpretation, withhold up to 3 points across the graphical-work rows.

Question 2 (20 points)

Item	Expected work	Points
2(a)	Correctly removes dominated strategies: A_5 by A_2, A_6 by A_3; B_3, B_4 by B_1 and B_5, B_6 by B_2.	2
2(a)	Forms the reduced 4×2 game and obtains $q_1 = q_2 = 1/2$ and $v = 6$ from the relevant A_2–A_3 intersection.	2
2(a)	Makes B_1 and B_2 indifferent and obtains $p_2 = 1/3$, $p_3 = 2/3$.	2
2(a)	Reports the full six-component strategies and checks that unused rows and columns cannot improve either player's payoff.	2
2(b)	Defines p_i and v_A and gives the objective $\max v_A$.	1
2(b)	Writes all six full-table payoff constraints in the correct direction. Award 0.5 per correct constraint, up to 3.	3
2(b)	Includes $\sum_i p_i = 1$, $p_i \geq 0$, and v_A unrestricted.	1
2(c)	Rewrites the payoff constraints as $v_A - \sum_i a_{ij} p_i \leq 0$ and assigns nonnegative dual variables q_j.	2
2(c)	Introduces unrestricted v_B for the primal equality and derives the dual objective $\min v_B$.	2
2(c)	Derives $\sum_j a_{ij} q_j \leq v_B$ for all six rows and $\sum_j q_j = 1$ from the free variable v_A.	2
2(c)	States that the resulting LP, including signs, is identical to the given Player B LP.	1
	Question 2 total	**20**

Question 3 (20 points)

Item	Expected work	Points
Setup	Defines a valid backward-DP value such as $G_i(s)$ and respects the "at least one day per remaining course" feasibility condition.	2
Course 4	Correct terminal-stage values $G_4(1, 2, 3, 4) = (6, 7, 9, 9)$.	3
Course 3	Shows the four maximizations and obtains $G_3(2, 3, 4, 5) = (8, 10, 13, 14)$.	5
Course 2	Shows the four maximizations and obtains $G_2(3, 4, 5, 6) = (13, 15, 18, 19)$.	5

Course 1	Evaluates $G_1(7) = \max\{22, 23, 21, 20\} = 23$.	3
Backtrack	Correctly backtracks to $(x_1, x_2, x_3, x_4) = (2, 1, 3, 1)$ and verifies $5 + 5 + 7 + 6 = 23$.	2
	Question 3 total	**20**

Question 4 (20 points)

Item	Expected work	Points
Setup	Correctly interprets (L, r), subtracts travel cost on a change, updates r, and excludes choosing L when $r = 2$.	2
Day 5	Correct terminal continuation and Day 5 values: $V_5(A, 1) = 7$, $V_5(A, 2) = 7$, $V_5(B, 1) = 10$, $V_5(B, 2) = 6$, $V_5(C, 1) = 8$, $V_5(C, 2) = 8$, with valid best actions.	3
Day 4	Correct values $(16, 14, 13, 13, 15, 15)$ for states $(A, 1), (A, 2), (B, 1), (B, 2), (C, 1), (C, 2)$ and valid best actions.	4
Day 3	Correct values $(21, 21, 23, 23, 25, 19)$ in the same state order and valid best actions.	4
Day 2	Correct values $(29, 29, 32, 30, 30, 30)$ in the same state order and valid best actions.	3
Day 1 and backtrack	Evaluates initial choices as $(37, 34, 32)$, selects A, and backtracks to the unique optimal sequence (A, B, C, B, B).	2
Final value	Correctly shows daily net contributions $(8, 6, 8, 5, 10)$ and total net profit 37.	2
	Question 4 total	**20**

Tie note: At off-path states, accept either valid tied action, including B or C at $V_5(C, 1)$, A or B at $V_4(B, 1)$, and A or C at $V_3(A, 1)$.

Question 5 (20 points)

Item	Expected work	Points
5(a)	Draws states $0, 1, 2, 3, 4$ with rightward rate λ through state 4 and leftward rate μ through state 0; recognizes arrivals at 4 are rejected.	2
5(b)	Uses local/global balance to obtain $\pi_{n+1} = \rho\pi_n$ with $\rho = 1/2$.	2
5(b)	Normalizes and reports $(\pi_0, \ldots, \pi_4) = (16, 8, 4, 2, 1)/31$.	2
5(c)	Utilization $1 - \pi_0 = 15/31$.	1
5(d)	States PASTA and obtains the loss fraction $\pi_4 = 1/31$. One point for PASTA/explanation and one for the value.	2
5(e)	Computes $\mathbb{E}[N] = \sum n\pi_n = 26/31$.	2
5(f)	Computes the effective arrival rate $\lambda_e = \lambda(1 - \pi_4) = 60/31$.	1
5(f)	Applies Little's Law to admitted jobs and obtains $\mathbb{E}[T] = (26/31)/(60/31) = 13/30$.	2
5(g)	Correct loss probabilities $1/511$ after capacity doubling and $1/341$ after speed doubling; concludes capacity doubling is better.	2
5(h)	Correct loss probabilities $6561/242461$ after capacity doubling and $81/6505$ after speed doubling; concludes speed doubling is better.	2

5(i)	Explains that added space is especially effective at light load, whereas at heavier load service capacity addresses the congestion bottleneck directly.	2
	Question 5 total	**20**

Question 6 (10 points)

Item	Expected work	Points
Law	States and correctly applies Little's Law $\mathbb{E}[N] = \lambda\mathbb{E}[T]$.	3
Substitution	Substitutes the provided $\mathbb{E}[T]$ and cancels λ.	2
Result	Obtains $\mathbb{E}[N] = 1/[A(1-\rho)^2]$.	3
Definitions	Retains the printed definitions of A and $\rho = \lambda/(\mu_1 + \mu_2) < 1$ without swapping μ_1 and μ_2.	2
	Question 6 total	**10**

Question 7 (10 points)

Item	Expected work	Points
Instance	Gives parameters with $\mu_2 > \mu_1$, $\lambda < \mu_1 + \mu_2$, and $\lambda < \mu_2$. Official example: $(5, 1, 10)$.	2
Two servers	Correctly computes ρ, A, and two-server $\mathbb{E}[N]$. For the official example: $\rho = 5/11$, $A = 163/66$, $\mathbb{E}[N] = 1331/978$.	3
Fast only	Uses the $M/M/1$ formula $\mathbb{E}[N] = \lambda/(\mu_2 - \lambda)$. The official example gives 1.	2
Comparison	Shows a strict improvement after disconnection; for the official example, $1 < 1331/978$.	1
Intuition	Explains how fast-server-first routing can send a job to a very slow server even when waiting briefly for the fast server would be better, while the fast server alone still has sufficient spare capacity.	2
	Question 7 total	**10**

Point-total check

$$20 + 20 + 20 + 20 + 20 + 10 + 10 = \boxed{120}.$$